

GROUP 8 WORKFLOW PROPOSAL

Meeting agenda:

- Create timeline and set smaller deadlines
- Ask tutor next week what they think of the plan
- Allocate member roles, or fix them.
 - MEMBER 1: Gia
 - MEMBER 2 Jamie
 - MEMBER 3 Rosa
 - MEMBER 4 Jisu
 - MEMBER 5 Skylar
 - MEMBER 6 Sabrina
- Feel free to change this proposal as much as you want

This approach is based on how tutorial 3 analyses data:

Raw order book → WAP → Log returns → Realised Volatility (30s bins)

then:

Train/val split (8 min / 2 min)

Then:

Regression (WLS) , HAV-RV , ARMA-GARCH

then:

Predict on validation set

then:

QLIKE + MSE → Boxplots

Data prep → model component → evaluation → communication piece

Member	Data responsibility	Model responsibility	Eval / comms piece	Load estimate
M1 Data lead + Pipeline	Build shared data module WAP + log returns RV computation (σ) 30-sec bucket splits EDA + sanity checks	Coordinate model inputs Ensure all models use same RV definition QA model outputs for data leakage / bugs	QLIKE + MSE functions Evaluation framework shared by all models Rolling window + one-shot scheme code	Heavy upfront Lighter in Phases 3–4. Critical path for whole team.
M2 Features + EDA	BidAskSpread feature Num orders, volume Feature selection EDA Correlation w/ RV Covariate final set	WLS regression model $\alpha = 0.88$ weights OLS vs WLS comparison Covariate: price, order, BidAskSpread	Regression results slide Coefficient plots Explain WLS to trader: "we weight recent data more"	Medium-heavy. Feature work is analytical, model is well-scoped.
M3 HAV-RV model	Realised quarticity RQ HAV-specific features Lagged RV windows (RV_t-1, 5-period mean) slider package usage	HAV-RV OLS + WLS $w_t = RV_{t-1} / \sqrt{RQ}$ Truncated to RV_t-1 + mean_5 (short series) β estimates + CIs	HAV results section Predicted vs actual RV plot for app Explain: "yesterday's vol predicts today's"	Medium. HAV is conceptually clean but quarticity needs care.
M4 ARMA-GARCH	Log returns time series per time_id bucket ACF/PACF diagnostics Order selection (AIC/BIC)	ARMA(1,1)-GARCH(1,1) arch / rugarch lib ugarchpath simulation 1000 paths × 30 steps Mean SD as RV pred	GARCH results section Simulated paths viz for app (fan chart) Explain volatility clustering to trader	Heaviest model track. Simulation loop can be slow — start early.
M5 App developer	Integrate all model outputs into app Live RV plot from shared data module Stock selector UI	No new model — wraps M2/M3/M4 outputs Side-by-side model comparison panel QLIKE/MSE table	Full Shiny/Dash app Trader UX: input a stock → see forecast App demo video walkthrough	Medium. Waits on other tracks but can scaffold UI early with mocks.
M6 Comms + theory	Order book theory docs (WAP, spread, log returns explainer) Fact-checks all written content	Naive benchmark models $\sigma^* = \sigma_{10min}$ (baseline) $\sigma^* = \sqrt{2} \cdot \sigma_{5min}$ Gives M2/M3/M4 a baseline to beat	Video script + record Infographic: what is vol, why it matters Options / Black-Scholes trader explainer	Medium. Naive benchmarks are quick to code; comms is ongoing.

